WORKED SOLUTION

STEP III 2012, Question 5

Pure • Rational Points on Conics

STEP III 2012, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s3-q5

The key idea

The identity in (i)(b) is a rotation in disguise: the point $(\cos\theta + \sqrt{m}\sin\theta,\ \sin\theta - \sqrt{m}\cos\theta)$ is what you get by rotating the point $(1, -\sqrt{m})$ through the angle θ, so it stays on the circle of radius $\sqrt{1+m}$. If $\cos\theta$ and $\sin\theta$ are both rational (such as $\frac{3}{5}$ and $\frac{4}{5}$, from (i)(a)), the rotation turns one "nice" point into another. Everything in part (ii) is the same trick, starting from a suitable integer point, and for the hyperbola we swap the rotation for its hyperbolic cousin.

Solution

Part (i)

(a) The point $(1,0)$ is an integer rational point on $x^2 + y^2 = 1$. For a non-integer one we use the $3,4,5$ triangle: $\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2 = \frac{9+16}{25} = 1$, so $\left(\frac{3}{5}, \frac{4}{5}\right)$ works, and neither coordinate is an integer.

(b) The point $(1,1)$ is an integer rational point on $x^2 + y^2 = 2$.

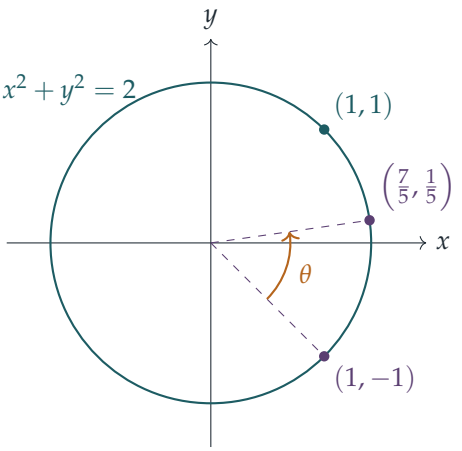
Expanding, the cross terms are $+2\sqrt{m}\sin\theta\cos\theta$ and $-2\sqrt{m}\sin\theta\cos\theta$, which cancel, so

$$\begin{aligned}(\cos\theta + \sqrt{m}\sin\theta)^2 + (\sin\theta - \sqrt{m}\cos\theta)^2 &= \cos^2\theta + m\sin^2\theta + \sin^2\theta + m\cos^2\theta \\ &= (1+m)(\cos^2\theta + \sin^2\theta) = 1 + m.\end{aligned}$$

Take $m = 1$, so the right-hand side is 2, and choose θ with $\cos\theta = \frac{3}{5}$, $\sin\theta = \frac{4}{5}$ (possible because $\left(\frac{3}{5}, \frac{4}{5}\right)$ lies on the unit circle). Then

$$x = \tfrac{3}{5} + \tfrac{4}{5} = \tfrac{7}{5}, \qquad y = \tfrac{4}{5} - \tfrac{3}{5} = \tfrac{1}{5}$$

satisfy $x^2 + y^2 = 2$; indeed $\frac{49+1}{25} = 2$. So $\left(\frac{7}{5}, \frac{1}{5}\right)$ is a non-integer rational point on $x^2 + y^2 = 2$.



The picture shows what happened: rotating $(1, -1)$ anticlockwise through θ, where $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$, lands on $\left(\frac{7}{5}, \frac{1}{5}\right)$.

Part (ii)

(a) Since $1^2 + (\sqrt{2})^2 = 3$, the point $(1, \sqrt{2})$, that is $p = 1$, $q = 0$, $r = 0$, $s = 1$, is an integer 2-rational point on $x^2 + y^2 = 3$.

For a non-integer one, use the identity from (i)(b) with $m = 2$, so that $1 + m = 3$, and again $\cos\theta = \frac{3}{5}$, $\sin\theta = \frac{4}{5}$:

$$x = \tfrac{3}{5} + \tfrac{4}{5}\sqrt{2}, \qquad y = \tfrac{4}{5} - \tfrac{3}{5}\sqrt{2}.$$

Here $p = \frac{3}{5}$, $q = \frac{4}{5}$, $r = \frac{4}{5}$, $s = -\frac{3}{5}$, none of which is an integer, and $x^2 + y^2 = 3$ by the identity. As a direct check, $x^2 = \frac{9+32+24\sqrt{2}}{25}$ and $y^2 = \frac{16+18-24\sqrt{2}}{25}$, which add to $\frac{75}{25} = 3$.

Ask yourself How can we reach 11, when the identity only gives $1 + m$ and m has to involve $\sqrt{2}$?

Scale the starting point. The identity came from rotating $(1, -\sqrt{m})$; we can just as well rotate any point $(a, -b\sqrt{2})$. The same cancellation of cross terms gives

$$(a\cos\theta + b\sqrt{2}\sin\theta)^2 + (a\sin\theta - b\sqrt{2}\cos\theta)^2 = (a^2 + 2b^2)(\cos^2\theta + \sin^2\theta) = a^2 + 2b^2.$$

So we need integers with $a^2 + 2b^2 = 11$, and $a = 3$, $b = 1$ works ($9 + 2 = 11$). In other words, we start from the integer 2-rational point $(3, \sqrt{2})$ on $x^2 + y^2 = 11$.

(b) With $a = 3$, $b = 1$, $\cos\theta = \frac{3}{5}$, $\sin\theta = \frac{4}{5}$:

$$x = \tfrac{9}{5} + \tfrac{4}{5}\sqrt{2}, \qquad y = \tfrac{12}{5} - \tfrac{3}{5}\sqrt{2}.$$

So $p = \frac{9}{5}$, $q = \frac{4}{5}$, $r = \frac{12}{5}$, $s = -\frac{3}{5}$, none an integer, and $x^2 + y^2 = 11$. Directly: $x^2 = \frac{81+32+72\sqrt{2}}{25}$ and $y^2 = \frac{144+18-72\sqrt{2}}{25}$, which add to $\frac{275}{25} = 11$.

(c) For the hyperbola we want a "rotation" that preserves $x^2 - y^2$ rather than $x^2 + y^2$. If $c^2 - s^2 = 1$ (think $c = \cosh t$, $s = \sinh t$), then

$$\begin{aligned}(ac + b\sqrt{2}\,s)^2 - (as + b\sqrt{2}\,c)^2 &= a^2c^2 + 2b^2s^2 - a^2s^2 - 2b^2c^2 \\ &= (a^2 - 2b^2)(c^2 - s^2) = a^2 - 2b^2,\end{aligned}$$

because the cross terms $2\sqrt{2}\,abcs$ cancel again. We need $a^2 - 2b^2 = 7$: take $a = 3$, $b = 1$ (so $(3, \sqrt{2})$ is an integer 2-rational point on the hyperbola).

Now we need rational c, s with $c^2 - s^2 = (c - s)(c + s) = 1$. Choosing $c + s = 2$ and $c - s = \frac{1}{2}$ gives $c = \frac{5}{4}$, $s = \frac{3}{4}$. Then

$$x = \tfrac{15}{4} + \tfrac{3}{4}\sqrt{2}, \qquad y = \tfrac{9}{4} + \tfrac{5}{4}\sqrt{2},$$

with $p = \frac{15}{4}$, $q = \frac{3}{4}$, $r = \frac{9}{4}$, $s = \frac{5}{4}$, none an integer. Directly: $x^2 = \frac{225+18+90\sqrt{2}}{16}$ and $y^2 = \frac{81+50+90\sqrt{2}}{16}$, so $x^2 - y^2 = \frac{112}{16} = 7$. This is a non-integer 2-rational point on $x^2 - y^2 = 7$.

Checks

Every point can be checked exactly with fractions, as done above, and numerically: for (ii)(c), $x \approx 4.81066$ and $y \approx 4.01777$, giving $x^2 - y^2 \approx 7.0000$; for (ii)(b), $x \approx 2.93137$ and $y \approx 1.55147$, giving $x^2 + y^2 \approx 11.0000$. Always finish by scanning p, q, r, s to confirm none is an integer.

Common slips

- Forgetting that 0 is an integer: a point like $\left(\frac{7}{5}\sqrt{2}, \dots\right)$ has $p = 0$, so it is not a non-integer 2-rational point.
- In (c), an innocent-looking choice can fail: $c = \frac{5}{3}$, $s = \frac{4}{3}$ also satisfies $c^2 - s^2 = 1$, but gives $x = 5 + \frac{4}{3}\sqrt{2}$, with $p = 5$ an integer. Check every coefficient.

Beyond STEP

Numbers of the form $p + q\sqrt{2}$ form a field, $\mathbb{Q}(\sqrt{2})$, and the question is really about finding points on conics over that field. The trick of rotating a known point by a rational angle is the start of a beautiful theory: once a conic has one rational point, lines through it parametrise all the others, and the rational points on $x^2 - 2y^2 = 1$ (Pell's equation) form a group under exactly the hyperbolic "rotation" used in (c). Davenport's *The Higher Arithmetic* covers Pell's equation and sums of squares, and Allenby's *Rings, Fields and Groups* introduces fields like $\mathbb{Q}(\sqrt{2})$.

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